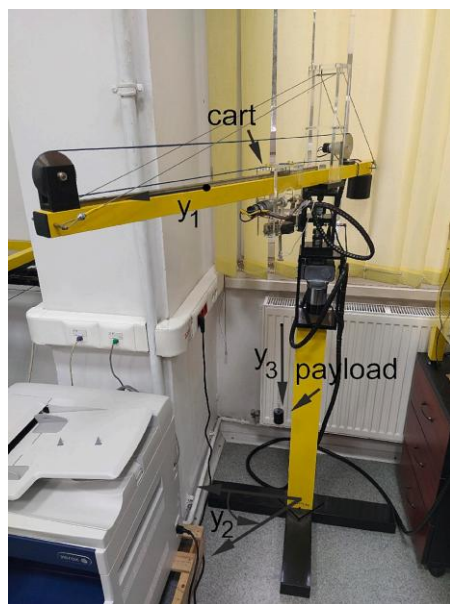




**A. Laboratory tower-crane safe control**

**EVL L2 | AM1**



**(a) Lifting of the concrete block**



**(b) Obstacle avoidance**



**(c) Transporting of the concrete block**

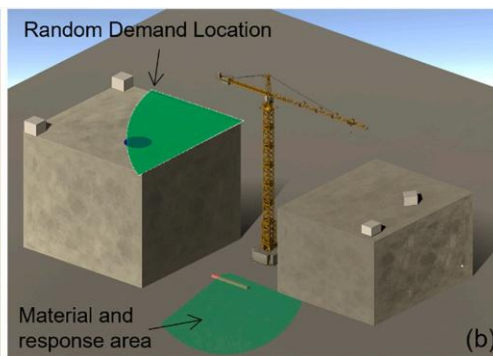
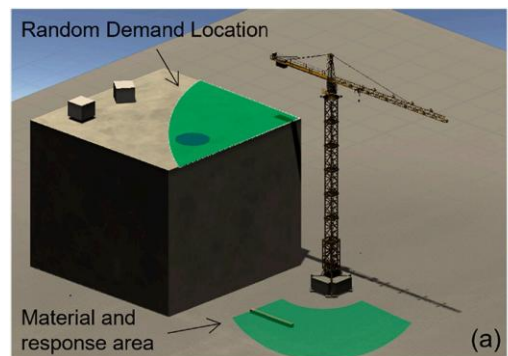
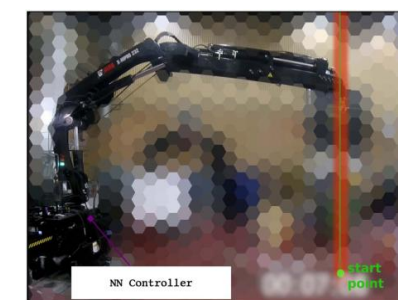


**(d) Reach the target position**

**B. Robotized mobile-crane placement and assistance control system**

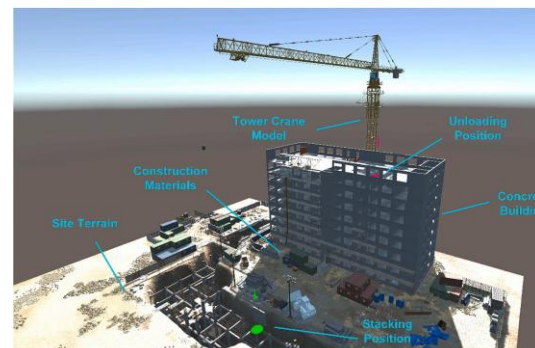
**(i) EVL L3 | AM2**

**(ii) EVL L2 | AM1**



**C. Tower-crane lift-path planning**

**(i) EVL L2 | AM0**



**(ii) EVL L0 | AM0**



**D. Double-pendulum crane trajectory planning**

**EVL L2 | AM1**

**EVL** = validation exposure; **AM** = runtime assurance